CLINICAL · APPROACH TO

Approach to facial weakness

A face that will not move is frightening for the patient and often treated too casually by the doctor. Most acute lower motor neuron palsies are idiopathic and recover, but Bell's palsy is a diagnosis of exclusion, and the approach is designed to find the patient who does not fit: the stroke that spares the forehead, the cholesteatoma behind the drum, the zoster vesicles in the concha, the parotid cancer under the angle of the jaw, and the slow, twitching palsy that is a tumour until imaging says otherwise. Whatever the cause, the eye comes first.

AT A GLANCE

- Forehead spared = upper motor neuron.
- Examine ear, parotid, other cranial nerves and skin for vesicles.
- Prednisolone within 72 hours; eye protection always.
- Slowly progressive or recurrent same-side palsy = tumour until imaged.

1 CHAPTER ONE Upper or lower?

The forehead decides.

1.1 The first distinction

The upper face receives fibres from both cerebral hemispheres, while the lower face receives fibres only from the opposite hemisphere. A **supranuclear (upper motor neuron)** lesion therefore weakens the lower face on the opposite side and spares the forehead, and emotional smiling may still move the mouth. A **lower motor neuron** lesion, at or beyond the facial nucleus in the pons, paralyses the whole of one side of the face, forehead included.

A forehead-sparing palsy of sudden onset is a stroke until proven otherwise and goes down the stroke pathway, not to the ENT clinic. A pontine lesion can weaken the face together with the sixth nerve, because the motor fibres loop around the abducens nucleus.

RED FLAGS IN FACIAL WEAKNESS

- **Forehead sparing, limb weakness, dysarthria or other brainstem signs:** stroke pathway now.
- **Slow onset over weeks, twitching or spasm before the weakness, or a palsy that recurs on the same side:** tumour until imaging says otherwise.
- **No recovery at all by three to four months, or an outcome worse than House–Brackmann III:** look again for a cause.
- **A discharging ear, cholesteatoma or acute otitis media:** an otological palsy that needs surgery or drainage.
- **A parotid mass or a facial skin cancer:** perineural malignancy.
- **Bilateral palsy:** Lyme disease, Guillain–Barré syndrome, sarcoidosis, leukaemia, HIV.

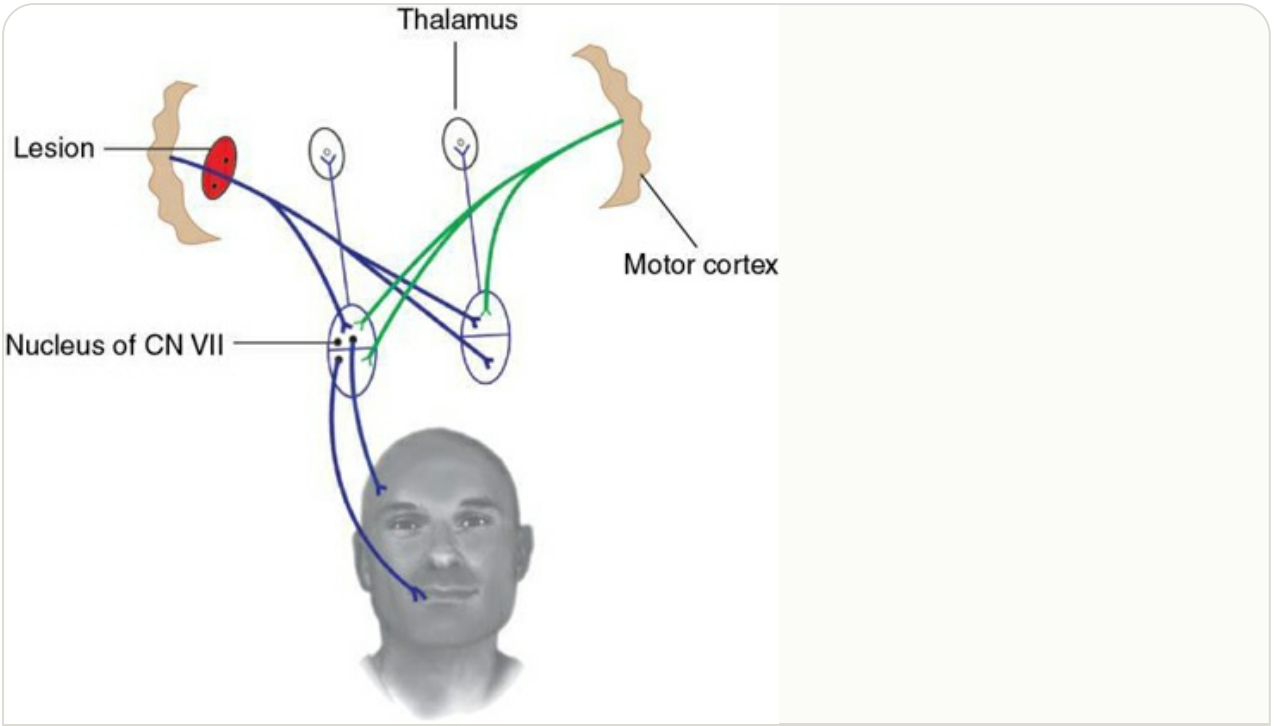


FIGURE 1 The forehead receives bilateral cortical innervation and is spared in supranuclear palsy.
Dhingra, fig. 14.1

CHAPTER TWO

2 History

Speed, sequence and company.

2.1 Onset and course

The single most useful question is how quickly the weakness developed. Bell's palsy reaches its worst within 72 hours, often within 24 to 48 hours, and is frequently preceded by pain behind the ear. A palsy that progresses over weeks, starts with twitching, or never begins to improve suggests a tumour of the nerve, the temporal bone or the parotid.

2.2 Associated symptoms

Ask about taste disturbance and hyperacusis, which place the lesion proximal to the chorda tympani and the nerve to stapedius but are common in Bell's palsy; about ear pain, a rash or blisters in the ear or mouth (Ramsay Hunt syndrome); about hearing loss, tinnitus and vertigo (zoster, cholesteatoma, a temporal bone fracture or a tumour of the internal auditory meatus or cerebellopontine angle); about discharge, previous ear disease and ear surgery; and about head injury and recent parotid or ear operations.

Ask also about tick bites, travel and rashes (Lyme disease), ascending weakness and numbness (Guillain–Barré syndrome), diabetes and hypertension, pregnancy, and previous episodes. Recurrent alternating palsies with facial swelling and a fissured tongue suggest Melkersson–Rosenthal syndrome.

For personal study. Follow local protocols and senior advice for patients.

3 CHAPTER THREE Examination

Grade, locate, find the cause.

3.1 The face and the eye

Look at the face at rest and with each movement: raising the brows, closing the eyes gently and tightly, smiling, puffing the cheeks and showing the teeth. Grade the weakness with the **House–Brackmann scale**, which runs from I (normal) to VI (total paralysis). The key break lies between III and IV: in grade III the eye still closes completely with effort, and from grade IV it no longer closes. Note **Bell's phenomenon** (the eye rolls up on attempted closure) and any exposure of the cornea, because an eye that does not close and does not roll up is at high risk.

HOUSE-BRACKMANN GRADING

GRADE	DESCRIPTION	EYE CLOSURE	FOREHEAD
I	Normal	Normal	Normal
II	Slight weakness	Complete, minimal effort	Moderate to good
III	Obvious, not disfiguring	Complete with full effort	Slight to moderate
IV	Disfiguring weakness	Incomplete	None
V	Barely perceptible movement	Incomplete	None
VI	Total paralysis	None	None

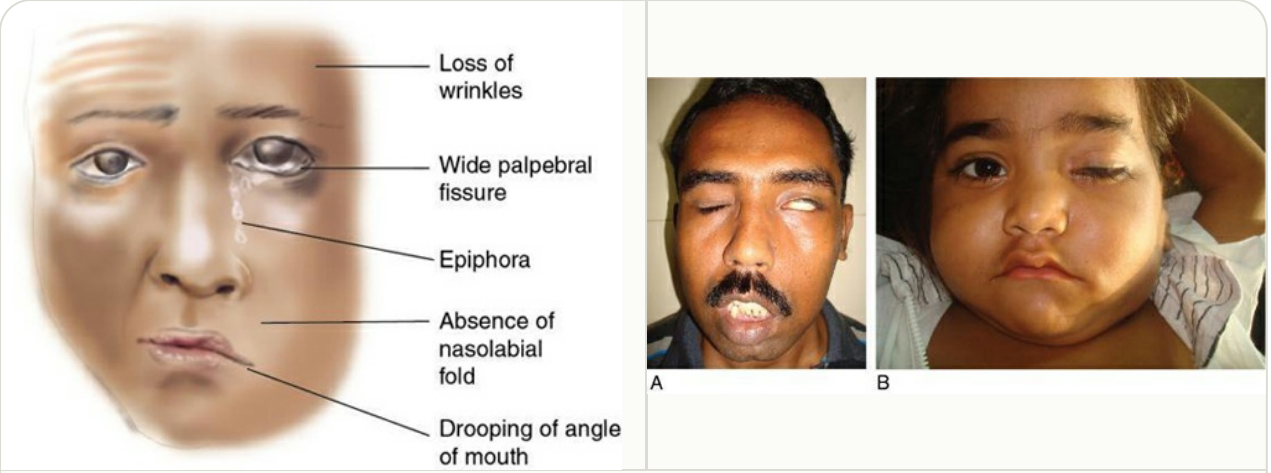


FIGURE 2 First, the features of a left lower motor neuron facial palsy. Second, Bell's palsy in an adult and a child.

Dhingra, fig. 14.7

3.2 Looking for a cause

Examine the ear under the microscope for cholesteatoma, otitis media, a mass behind the drum or blood, and look for **vesicles** in the concha, the canal, the soft palate and the anterior tongue. Test hearing with tuning forks. Palpate the parotid and the neck, and look at the facial skin and scalp for a cancer that may spread along the nerve. Examine the other cranial nerves, the limbs and the gait.

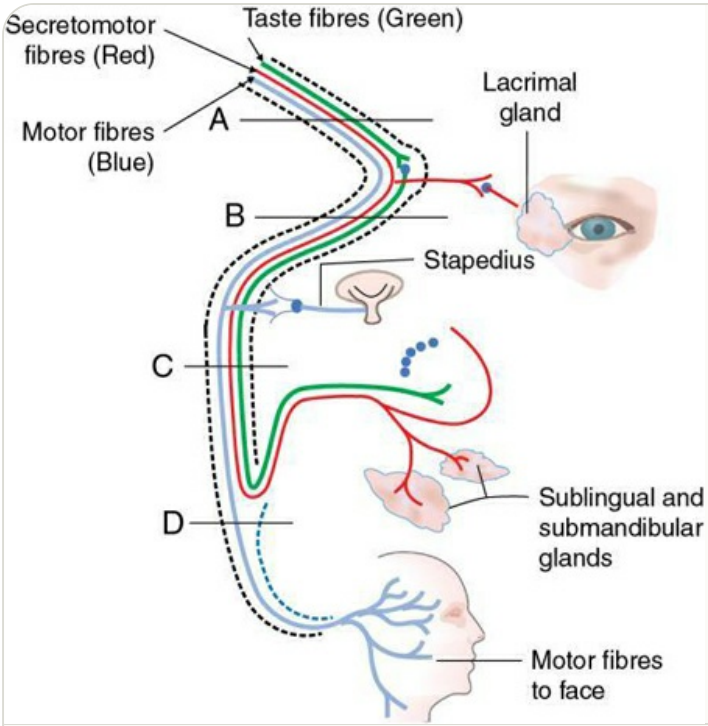


FIGURE 3 Topographical localisation of facial nerve lesions.
Dhingra, fig. 14.12

4 CHAPTER FOUR Causes

By site and by speed.

4.1 The common causes

Bell's palsy accounts for about 70% of acute lower motor neuron palsies. It affects about 30 people per 100,000 per year, peaks around 40 years and recurs in about 7%. **Ramsay Hunt syndrome**, caused by reactivation of varicella zoster in the geniculate ganglion, is the second commonest and carries a worse prognosis, with only about 20% recovering completely. The vesicles may appear late or not at all (*zoster sine herpete*), so a "Bell's palsy" with severe pain, hearing loss or vertigo deserves antiviral treatment.

Otological causes include acute otitis media in children, where the canal is often dehiscent and the prognosis is excellent with antibiotics and a myringotomy; cholesteatoma, which causes a palsy in under 1% of cases but in up to 50% of petrous apex disease and needs surgery; and necrotising otitis externa, with pain,

discharge and progressive cranial neuropathies in an elderly diabetic. **Trauma** includes temporal bone fractures, especially those crossing the otic capsule, and surgical injury in the middle ear, the mastoid, the cerebellopontine angle and the parotid.

Tumours may arise in the nerve (schwannoma and haemangioma), press on it (paraganglioma, meningioma, vestibular schwannoma) or invade it (parotid cancer, cutaneous squamous cell carcinoma with perineural spread). **Neurological and systemic causes** include stroke, Guillain–Barré syndrome, Lyme disease, sarcoidosis and, in children, congenital palsies such as Möbius syndrome and CHARGE.

WHERE THE LESION LIES

SITE	CLUES	EXAMPLES
Supranuclear	Forehead spared; limb or speech signs	Stroke, tumour
Pons	Sixth nerve palsy; long tract signs	Infarct, demyelination
Cerebellopontine angle and meatus	Hearing loss, tinnitus, imbalance; other cranial nerves	Vestibular schwannoma, meningioma
Temporal bone	Ear disease; hearing loss; vertigo; vesicles	Bell's palsy, Ramsay Hunt syndrome, cholesteatoma, fracture
Parotid and face	Mass; selective branch weakness; skin cancer	Parotid malignancy, surgery, trauma



FIGURE 4 Course of the facial nerve: meatal, labyrinthine, tympanic and mastoid segments, and its branches.

Dhingra, fig. 14.2

5 CHAPTER FIVE Investigation

What and when.

5.1 Tests

A typical Bell's palsy in an adult with a normal ear and no other neurological findings needs no tests at presentation, but a pure tone audiogram is useful in every patient and essential if there is hearing loss or vertigo. Blood glucose is checked when diabetes is possible, Lyme serology in endemic areas or with a tick bite, and a PCR swab of vesicles confirms zoster.

Imaging is needed for any atypical feature: slow onset, twitching, recurrence on the same side, no recovery by three to four months, other cranial neuropathies, hearing loss, ear disease or a parotid mass. Gadolinium-enhanced MRI covers the nerve from the brainstem to the parotid, and high-resolution CT shows the bony canal, fractures and cholesteatoma.

Electroneuronography compares the compound muscle action potential on the two sides in a complete palsy between day 3 and day 21. More than 90% degeneration predicts a poor recovery and is used in some centres to select patients after trauma for exploration. **Electromyography** after three weeks shows fibrillation potentials (degeneration with viable muscle) and polyphasic potentials (early reinnervation); electrical silence after a long palsy means the muscle is no longer viable.

6 CHAPTER SIX Pathway

From weak face to plan.

THE WEAK FACE

1. Forehead moving?
   - → **spared**
     stroke pathway, neurology
   - → **paralysed**
     lower motor neuron, step 2
2. Eye care now
   - → **incomplete closure or poor Bell's phenomenon**
     lubricating drops by day, ointment and taping at night, ophthalmology if exposed

3 **Ear, parotid and nerves examined**

- → **vesicles, pain, hearing loss or vertigo**
  Ramsay Hunt — steroid and antiviral
- → **cholesteatoma or otitis media**
  CT and surgery or drainage
- → **parotid mass or skin cancer**
  imaging and fine needle aspiration
- → **other neurology or bilateral**
  neurology, Lyme serology

4 **Typical sudden palsy within 72 hours**

- → Bell's palsy — prednisolone within 72 hours, eye care, review

5 **Not recovering by three to four months, slow, twitching or recurrent**

- → MRI with gadolinium and CT of the temporal bone
- → refer to a facial nerve clinic

CHAPTER SEVEN

7 Management

Principles for each group.

7.1 Bell's palsy

Prednisolone started within 72 hours improves the chance of full recovery; a typical adult course is 50 to 60 mg daily for 5 to 10 days, with or without a taper. Antivirals alone do not help, and adding them to steroids offers at most a small benefit, so they are reserved for severe palsy or suspected zoster. Every patient with incomplete closure needs eye protection. About 85% begin to recover within three weeks and most recover fully; complete palsy at onset, age over 60, severe pain, diabetes, hypertension and no recovery by three weeks predict a worse outcome.

Surgical decompression of the labyrinthine segment, the narrowest part of the canal at about 0.6 mm, remains controversial.

7.2 Ramsay Hunt syndrome

Treat with steroids and an antiviral such as aciclovir or valaciclovir, protect the eye, manage pain and vertigo, and refer for facial therapy early.

7.3 Trauma

An immediate complete palsy after a temporal bone fracture, with CT evidence of disruption of the canal or more than 90% degeneration on electroneuronography, is considered for exploration and repair. A delayed or incomplete palsy almost always recovers with steroids and eye care. A nerve known to have been cut at surgery is repaired at once.

7.4 Long-term weakness

Patients left with weakness need facial therapy, protection of the eye with gold or platinum weights in the upper lid and lower lid tightening, and botulinum toxin for synkinesis. Reanimation depends on whether the muscles are still viable, which is unlikely after 18 to 24 months: nerve repair, grafting or nerve transfer (masseteric or hypoglossal) while they are, and muscle transposition or free gracilis transfer once they are not.

8 CHAPTER EIGHT For the examination

What the examiner wants to hear.

8.1 Viva essentials

> WHAT THE EXAMINER IS LOOKING FOR
>
> - **Upper versus lower motor neuron:** the forehead and why.

- **Bell's palsy as a diagnosis of exclusion:** the features that make you look for another cause.
- **House–Brackmann grading** and the importance of eye closure.
- **Ramsay Hunt syndrome:** zoster sine herpete, antivirals, prognosis.
- **Electroneuronography and electromyography:** timing and meaning.
- **Steroids within 72 hours;** the limited role of antivirals.

8.2 Common pitfalls

COMMON PITFALLS

- Labelling a slowly progressive or twitching palsy as Bell's palsy.
- Forgetting the eye: an unprotected cornea can be lost while the face recovers.
- Never looking in the ear or feeling the parotid.
- Not reviewing the patient: a palsy that has not begun to recover by three to four months needs imaging.

Written for this app from Oxford Textbook of Otolaryngology ch. 10, Oxford Specialist Handbook ch. 3, Oxford Handbook of ENT ch. 3, Oxford Specialist Handbook ch. 17, Dhingra ch. 8, K. J. Lee ch. 19. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

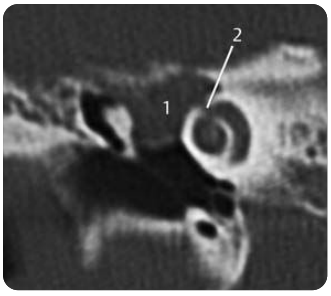
On imaging

The imaging of this topic: key findings, differentials and representative scans.

FACIAL NERVE · TUMOURS

Schwannoma of the Facial Nerve

- CT and MRI are complementary in the evaluation of the extension of the schwannoma to the middle or posterior cranial fossa as well as to the inner ear structures. Compared with hemangiomas, schwannomas of the facial nerve can be quite large before symptoms occur.
- Surgical removal with facial nerve grafting is only considered in (near) total paralysis of the facial nerve or when expansion of the lesion would lead to the damage of surrounding critical structures.

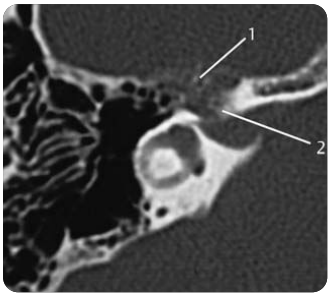


CT, coronal. CT demonstrates an opacification (1) in the medial and anterior epitympanum, suggestive of an expanding mass with destruction …

FACIAL NERVE · TUMOURS

Hemangioma of the Geniculate Ganglion

- Signs of infiltration or expansion. An MRI may be necessary to exclude or evaluate intracranial involvement.
- Most tumors, such as schwannoma, demonstrate more sharply defined enhancement.
- Outlines are often poorly defined in hemangiomas. A characteristic salt and pepper configuration is due to accumulation of methemoglobin and flow voids from the increased vascularization. Ossifying hemangiomas might demonstrate bone spicules.
- Hemangiomas of the facial nerve become symptomatic at an earlier stage compared with schwannomas, which might be quite large before symptoms occur.



CT, axial. There is a spiculated area of bone destruction or a mass with calcifications in the region of the geniculate ganglion (1), just distal to…

SALIVARY GLANDS · TUMOURS

Malignancies of the Parotid Gland

IMAGING FINDINGS

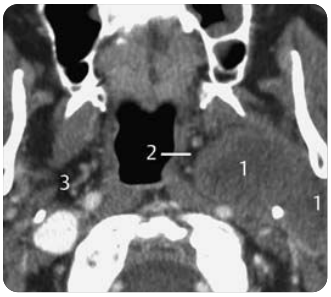
- Ill-defined or infiltrative margins
- Low or intermediate T2, low ADC
- Facial nerve: enhancement through the stylomastoid foramen

TOP DIFFERENTIALS

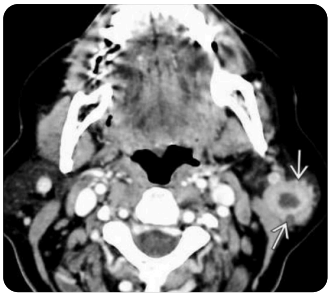
- Pleomorphic adenoma
- Warthin tumour
- Metastatic node from skin cancer; lymphoma

CHECKLIST

- Perineural spread to the mastoid segment and beyond
- Nodes in the parotid and level II



Slight swelling of the left parotid region and deep pain.
Dunnebier, Imaging for Otolaryngologists, Fig. 9.30 a



Axial CECT reveals a well-defined mass in the superficial lobe of the parotid gland. It has a thick rind of peripheral enhancement and is centrally cystic …

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 98.

TEMPORAL BONE & SKULL BASE · TRAUMA

Skull Base Fractures

IMAGING FINDINGS

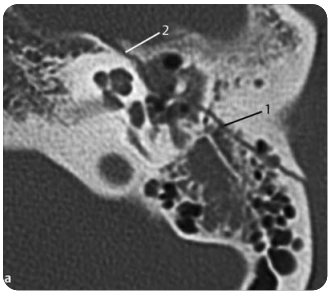
- Fracture line through the temporal bone or skull base; air in the middle ear or cranium
- Otic capsule sparing (most) or violating
- Fluid in the middle ear and mastoid

TOP DIFFERENTIALS

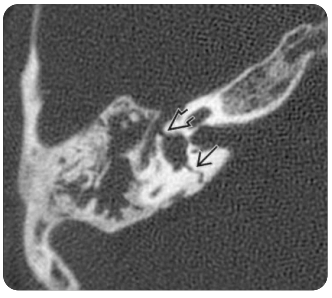
- Sutures, fissures and small canals (pseudofractures)
- Vascular grooves

CHECKLIST

- Ossicles, facial canal (geniculate), tegmen and CSF leak
- Carotid canal involved: CT angiography



CT, axial. Patient with head trauma showing a fracture line through the mastoid (1) extending just anterior to the cochlea (2). Luxation of the…



Axial bone CT in a 14-year-old following severe head injury shows a medial transverse, otic capsule-violating fracture traversing the vestibule, with…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 968.